



1. Helpdesk Services

This chapter defines Helpdesk Services. The API URI is defined as:

`https://[FQDN]/helpdesk/admin/api/[version]`

1.1. Get Device Set Information

Purpose	This interface returns the current Device set configuration in the system
Path URL	/device-set/{setId}/info
HTTP Method	GET
Version	1.0
Request Headers	Authorization: Bearer JWT_VALUE Content-Type: application/json
Request Content	None
Request Path Parameter	setId – Unique identification of Product Set
HTTP Response Codes	200, 400, 401, 403
Response Headers	Content-Type: application/json
Response Content	See Response Content

1.1.1. Response Content

Property	Type	Mand	Constraints	Description
setId	String	Yes	Limit 100	Uniquely identify the device set
productName	String	Yes	Limit 100	Uniquely identify the product
productModel	String	Yes	Limit 100	Model of the product
latestConfiguration	<i>Error! Reference source not found.</i>	Yes		Indicates the current configuration of Device set



Property	Type	Mand	Constraints	Description
tags	Map<String,String>	No	>=0	Key-> value pairs of device set Tags Supported value dataTypes {STRING, INTEGER, DOUBLE, BOOLEAN}

1.1.2. Example Request

GET https://[FQDN]/helpdesk/admin/api/1.0/device-set/JVC000VQ5KQSU9XAY812/info

1.1.3. Example Response

```
{
  "latestConfiguration": {
    "designName": "Design1-812",
    "elements": [
      {
        "address": "Address-812",
        "serialNumber": "SerialNo-812",
        "softwareId": "SW1Start-812",
        "name": "ECUName-812",
        "typeId": "TypeId-812",
        "imageDigest": "Tkqdx7arURjQ8COQfiEfcuyZfsBil66GAX0svgPeWb0="
      }
    ],
    "mapped": true,
    "clientTime": "2023-06-18T21:15:25Z[UTC]",
    "reportedTime": "2023-06-18T21:15:25Z[UTC]"
  },
  "productModel": "ProductModel-812",
  "setId": "JVC000VQ5KQSU9XAY812",
  "productName": "ProductName-812",
  "tags": {
    "tag1": "1000",
    "tag2": "false"
  }
}
```

1.2. Get Device set Configuration History

Purpose	This interface returns paginated list of configurations reported by device set in configMatch
Path URL	/device-set/{setId}/config-history



HTTP Method	GET
Version	1.0
Request Headers	Authorization: Bearer JWT_VALUE Content-Type: application/json
Request Path Parameter	setId – Unique identification of Product Set
Request Content	None
HTTP Response Codes	200, 400, 401, 403
Response Headers	Content-Type: application/json
Response Content	See Error! Reference source not found.

1.2.1. Response Content

The response contains paginated list of Error! Reference source not found. history

1.2.2. Example Request

GET https://[FQDN]/helpdesk/admin/api/1.0/device-set/JVC000VQ5KQSU9XAY812/config-history

1.2.3. Example Response

```
{
  "limit": 100,
  "totalPages": 1,
  "sort": "clientTime",
  "dir": "DESC",
  "totalCount": 3,
  "results": [
    {
      "elements": [
        {
          "serialNumber": "SerialNo-812",
          "typeId": "TypeId-812"
        },
        {
          "serialNumber": "test-SerialNo-812",
          "typeId": "TypeId-812"
        }
      ]
    },
    "mapped": false
  ]
}
```



```
    },
    {
      "designName": "Design1-812",
      "elements": [
        {
          "address": "Address-812",
          "serialNumber": "SerialNo-812",
          "softwareId": "SW1Start-812",
          "name": "ECUName-812",
          "typeId": "TypeId-812",
          "imageDigest": "Tkqdx7arURjQ8COQfiEfcuyZfsBil66GAX0svgPeWb0="
        }
      ],
      "mapped": true,
      "clientTime": "2023-06-18T21:15:25Z[UTC]",
      "reportedTime": "2023-06-18T21:15:25Z[UTC]"
    }
  ]
}
```

1.3. Get Event Notifications

Purpose	This interface returns paginated list of event notification sent by device sets.
Path URL	/device-set/{setId}/event-notifications? Search Query Parameters Error! Reference source not found.
HTTP Method	GET
Version	1.0
Request Headers	Authorization: Bearer JWT_VALUE Content-Type: application/json
Request Path Parameter	setId – Unique identification of Product Set
Request Content	None
HTTP Response Codes	200, 400, 401, 403
Response Headers	Content-Type: application/json
Response Content	See Response Content



1.3.1. Search Query Parameters

Property	Type	Description
status	String	Filter by notifications status. Allowed fields are {SUCCESS, FAILURE, IN_PROGRESS}
dateFrom	ISO 8601 Date-Time	Begin of notification event time range.
dateTo	ISO 8601 Date-Time	End of notification event time range.

1.3.2. Response Content

The response contains paginated list of **EventNotification**

1.3.3. EventNotification

Property	Type	Mand	Constraints	Description
setId	String	Yes		
campaignName	String	Yes		Name of an campaign
campaignDesc	String	Yes		Campaign description
packageName	String	Yes		Package name
packageDesc		Yes		Package description
attemptId	String	Yes		Single campaign events flow identification
noOfRetries	int	No		
eventType	String	Yes		
status	String	Yes		
eventTime	Date	Yes		
startTime	Date	No		
totalBytes	Long	No		
bytesDownloaded	Long	No		



Property	Type	Mand	Constraints	Description
address	String	Yes	Limit 100	The physical address of element
typeId	String	Yes	Limit 100	Type of element
softwareId	String	Yes	Limit 100	software version
serialNumber	String	Yes	Limit 100	
errorCodes	List<EventErrors>	No	Size >= 0	Contains codea and description of event errors
eventTransport	String			{wifi,cellular}
valid	Boolean	Yes		Based on some fields availability in notifications, the valid flag set to tru of false

1.3.4. EventErrors

Property	Type	Mandatory	Description
code	int	Yes	
description	String	Yes	

1.3.5. Example Request

GET https://[FQDN]/helpdesk/admin/api/1.0/device-set/JVC000VQ5KQSU9XAY812/event-notifications

1.3.6. Example Response

```
{
  "limit": 100,
  "totalPages": 1,
  "sort": "eventTime",
  "dir": "DESC",
  "totalCount": 10,
  "results": [
    {
      "address": "Address-812",
      "campaignDesc": "The Campaign Description",
      "serialNumber": "SerialNo-812",
      "errorCodes": [

```



```
"eventType": "installation",
"noOfRetries": 1,
"valid": true,
"packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
"bytesDownloaded": 1048576,
"softwareId": "SW1End-812",
"eventTime": "2023-06-18T21:15:46.713Z[UTC]",
"setId": "JVC000VQ5KQSU9XAY812",
"totalBytes": 1048576,
"startTime": "2023-06-18T21:15:46.713Z[UTC]",
"typeId": "TypeId-812",
"packageName": "Package-812",
"campaignName": "Campaign-812",
"attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
"status": "success"
},
{
  "address": "Address-812",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo-812",
  "errorCodes": [
    {
      "code": "404",
      "description": "not found"
    }
  ],
  "eventType": "installation",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 0,
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.649Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 0,
  "startTime": "2023-06-18T21:15:46.649Z[UTC]",
  "typeId": "TypeId-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
  "attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
  "status": "failure"
},
{
  "address": "Address-812",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo-812",
  "errorCodes": [
  ],
  "eventType": "installation",
```



```
"noOfRetries": 0,
"valid": true,
"packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
"bytesDownloaded": 0,
"softwareId": "SW1End-812",
"eventTime": "2023-06-18T21:15:46.589Z[UTC]",
"setId": "JVC000VQ5KQSU9XAY812",
"totalBytes": 0,
"typeId": "TypeId-812",
"packageName": "Package-812",
"campaignName": "Campaign-812",
"attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
"status": "started"
},
{
  "campaignDesc": "The Campaign Description",
  "errorCodes": [
  ],
  "eventType": "installation",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 0,
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.525Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 0,
  "typeId": "TypeId-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
  "attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
  "status": "accepted"
},
{
  "campaignDesc": "The Campaign Description",
  "errorCodes": [
  ],
  "eventType": "installation",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 0,
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.453Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 0,
  "typeId": "TypeId-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
```




```
"attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
"status": "declined"
},
{
  "address": "Address-812",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo-812",
  "errorCodes": [
  ],
  "eventType": "download",
  "noOfRetries": 1,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 1048576,
  "eventTransport": "wifi",
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.382Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 1048576,
  "startTime": "2023-06-18T21:15:46.382Z[UTC]",
  "typeId": "TypeId-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
  "attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
  "status": "success"
},
{
  "address": "Address-812",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo-812",
  "errorCodes": [
    {
      "code": "404",
      "description": "not found"
    },
    {
      "code": "415",
      "description": "not supported type"
    }
  ],
  "eventType": "download",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 1048500,
  "eventTransport": "wifi",
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.298Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
```



```
"totalBytes": 1048576,
"startTime": "2023-06-18T21:15:46.298Z[UTC]",
"typeld": "Typeld-812",
"packageName": "Package-812",
"campaignName": "Campaign-812",
"attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
"status": "failure"
},
{
  "address": "Address-812",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo-812",
  "errorCodes": [
  ],
  "eventType": "download",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 0,
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.221Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 0,
  "typeld": "Typeld-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
  "attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
  "status": "started"
},
{
  "campaignDesc": "The Campaign Description",
  "errorCodes": [
  ],
  "eventType": "download",
  "noOfRetries": 0,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 0,
  "softwareId": "SW1End-812",
  "eventTime": "2023-06-18T21:15:46.135Z[UTC]",
  "setId": "JVC000VQ5KQSU9XAY812",
  "totalBytes": 0,
  "typeld": "Typeld-812",
  "packageName": "Package-812",
  "campaignName": "Campaign-812",
  "attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
  "status": "accepted"
},
{

```



```
"campaignDesc": "The Campaign Description",
"errorCodes": [
],
"eventType": "download",
"noOfRetries": 0,
"valid": true,
"packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
"bytesDownloaded": 0,
"softwareId": "SW1End-812",
"eventTime": "2023-06-18T21:15:26.031Z[UTC]",
"setId": "JVC000VQ5KQSU9XAY812",
"totalBytes": 0,
"typeId": "TypeId-812",
"packageName": "Package-812",
"campaignName": "Campaign-812",
"attemptId": "745d531a-2442-4961-9765-353e82e9bad4",
"status": "declined"
}
]
```

1.3.7. Example Request

GET [https://\[FQDN\]/helpdesk/admin/api/1.0/device-set/Q731R59QDB7M9K73H476/event-notifications?dateFrom=2023-01-01&dateTo=2023-09-24T23:59:59&status=SUCCESS](https://[FQDN]/helpdesk/admin/api/1.0/device-set/Q731R59QDB7M9K73H476/event-notifications?dateFrom=2023-01-01&dateTo=2023-09-24T23:59:59&status=SUCCESS)

1.3.8. Example Response

```
{
  "limit": 100,
  "totalPages": 1,
  "sort": "eventTime",
  "dir": "DESC",
  "totalCount": 2,
  "results": [
    {
      "address": "Address1-476",
      "campaignDesc": "The Campaign Description",
      "serialNumber": "SerialNo1-476",
      "errorCodes": [
      ],
      "eventType": "installation",
      "noOfRetries": 1,
      "valid": true,
      "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
      "bytesDownloaded": 1048576,
      "softwareId": "SW1End-476",
      "eventTime": "2023-09-24T22:25:06.929Z[UTC]",
    }
  ]
}
```



```
{
  "setId": "Q731R59QDB7M9K73H476",
  "totalBytes": 1048576,
  "startTime": "2023-09-24T22:25:06.929Z[UTC]",
  "typeId": "TypeId1-476",
  "packageName": "Package1-476",
  "campaignName": "Campaign-476",
  "attemptId": "34ab745e-3042-454a-a5a8-0e228c630673",
  "status": "success"
},
{
  "address": "Address1-476",
  "campaignDesc": "The Campaign Description",
  "serialNumber": "SerialNo1-476",
  "errorCodes": [
  ],
  "eventType": "download",
  "noOfRetries": 1,
  "valid": true,
  "packageDesc": "This is a firmware upgrade which addresses security vulnerability.",
  "bytesDownloaded": 1048576,
  "eventTransport": "wifi",
  "softwareId": "SW1End-476",
  "eventTime": "2023-09-24T22:25:06.457Z[UTC]",
  "setId": "Q731R59QDB7M9K73H476",
  "totalBytes": 1048576,
  "startTime": "2023-09-24T22:25:06.457Z[UTC]",
  "typeId": "TypeId1-476",
  "packageName": "Package1-476",
  "campaignName": "Campaign-476",
  "attemptId": "dc73dc0a-6d6c-4c7d-9986-568451ce1e4e",
  "status": "success"
}
]
```

1.4. Create Helpdesk Note

Purpose	This interface adds helpdesk notes to a given device set
Path URL	/device-set/{setId}/note
HTTP Method	POST
Version	1.0
Request Headers	Authorization: Bearer JWT_VALUE Content-Type: application/json



Request Path Parameter	setId – Unique identification of Device Set
Request Content	See Request Content
HTTP Response Codes	201, 400, 401, 403
Response Headers	Content-Type: application/json
Response Content	None

1.4.1. Request Content

The request contains a text to store helpdesk note with limit of 1000 characters.

1.4.2. Example Request

POST https://[FQDN]/helpdesk/admin/api/1.0/device-set/JVC000VQ5KQSU9XAY812/note

Content

Update failed due to connection.

1.4.3. Example Response

201 OK

1.5. Get Device set Helpdesk Notes

Purpose	This interface returns paginated list of helpdesk notes for a given Device set.
Path URL	/device-set/{setId}/notes
HTTP Method	GET
Version	1.0
Request Headers	Authorization: Bearer JWT_VALUE Content-Type: application/json
Request Path Parameter	setId – Unique identification of Device Set
Request Content	None
HTTP Response Codes	200, 400, 401, 403



Response Headers	Content-Type: application/json
Response Content	See Response Content

1.5.1. Response Content

The response contains paginated list of **Helpdesk Note**.

1.5.2. Helpdesk Note

Property	Type	Mand	Constraints	Description
id	Long	Yes		System generated unique identification of each helpdesk note.
note	String	Yes	Limit 1000	helpdesk note description
createdBy	String	Yes	Limit 100	Identification of the user who added the note
creationDate	Long	Yes		The helpdesk note creation date

1.5.3. Example Request

GET https://[FQDN]/helpdesk/admin/api/1.0/device-set/JVC000VQ5KQSU9XAY812/notes

1.5.4. Example Response

```
{
  "totalPages": 1,
  "limit": 100,
  "totalCount": 1,
  "results": [
    {
      "note": "Update failed due to connection.",
      "createdBy": "dfc923cc-ce4d-4836-affb-dc8bdfe71b80",
      "id": 18,
      "creationDate": 1634152887733
    }
  ]
}
```